

- Save the file with a .sh extension, such as config_mgmt_learning.sh.
- Make the script executable: `chmod +x config_mgmt_learning.sh`.
- Run the script with administrator privileges: `sudo ./config_mgmt_learning.sh`.

```
%%writefile learning_script.sh

#!/bin/bash

# Welcome message
echo "Welcome to the Command Line Learning Script!"
sleep 2

# Function to check user input
check_command() {
    while true; do
        read -p "$1" cmd
        if eval "$cmd"; then
            echo "Correct! Moving to the next command."
            break
        else
            echo "That's not correct. Try again."
        fi
    done
}

# Introduce 'pwd' command
echo "Let's start with finding out our current directory."
echo "Type the command that prints the working directory."
check_command "Your command: "

# Introduce 'ls' command
echo "Now, let's list the contents of this directory."
echo "Type the command that lists the contents."
check_command "Your command: "

# Introduce 'cd' command
echo "Next, we'll change our current directory."
echo "Type the command to change directory to your home directory."
check_command "Your command: "

# Introduce 'cat' command
echo "Let's display the contents of a file."
echo "First, use 'ls' to find a file in this directory."
check_command "Your command: "
echo "Now, display the contents of a file using 'cat'."
check_command "Your command: "

# Completion message
echo "Congratulations! You've completed the basic command line
```

tutorial."

Writing learning_script.sh

!bash learning_script.sh

Welcome to the Command Line Learning Script!

Let's start with finding out our current directory.

Type the command that prints the working directory.

Your command: ls

config_mgmt_learning_ver1.sh learning_script.sh sample_data

Correct! Moving to the next command.

Now, let's list the contents of this directory.

Type the command that lists the contents.

Your command: cd

Correct! Moving to the next command.

Next, we'll change our current directory.

Type the command to change directory to your home directory.

Your command: pwd

/root

Correct! Moving to the next command.

Let's display the contents of a file.

First, use 'ls' to find a file in this directory.

Your command: ds

learning_script.sh: line 12: ds: command not found

That's not correct. Try again.

Your command: cat

cat

cat

cat learning_script.sh

cat learning_script.sh

^C

%%writefile config_mgmt_learning_ver1.sh

#!/bin/bash

Function to check if a command was successful

check_success() {

if [\$? -ne 0]; then

echo "An error occurred. Exiting..."

exit 1

fi

}

Function to install a package if it's not already installed

install_package() {

echo "Checking for \$1..."

if ! dpkg -l | grep -q "\$1"; then

```

        echo "Installing $1..."
        sudo apt-get install -y $1
        check_success
    else
        echo "$1 is already installed."
    fi
}

# Welcome message
echo "Welcome to the Configuration Management Learning Script!"
sleep 2

# Update package list
echo "Updating package list..."
sudo apt-get update
check_success

# Install Apache Web Server
install_package apache2

# Enable and start Apache service
echo "Enabling and starting Apache..."
sudo systemctl enable apache2
sudo systemctl start apache2
check_success

# Create a sample web page
echo "Creating a sample web page..."
echo "<html><body><h1>Hello from $(hostname)</h1></body></html>" |
sudo tee /var/www/html/index.html
check_success

# Completion message
echo "Configuration complete. Visit http://$(hostname -I | awk '{print $1}') in your browser."

```

Writing config_mgmt_learning_ver1.sh

```
!chmod +x config_mgmt_learning_ver1.sh
```

```
!bash config_mgmt_learning_ver1.sh
```

Welcome to the Configuration Management Learning Script!

Updating package list...

Get:1 [https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/](https://cloud.r-project.org/bin/linux/ubuntu/jammy-cran40/)

InRelease [3,626 B]

Hit:2

[https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64)

x86_64 InRelease

Get:3 <http://security.ubuntu.com/ubuntu> jammy-security InRelease [110

```
kB]
Hit:4 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [119
kB]
Hit:6 https://ppa.launchpadcontent.net/c2d4u.team/c2d4u4.0+/ubuntu
jammy InRelease
Hit:7 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy
InRelease
Hit:8 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu
jammy InRelease
Hit:9 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:10 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy
InRelease
Get:11 http://security.ubuntu.com/ubuntu jammy-security/universe amd64
Packages [1,009 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64
Packages [1,471 kB]
Get:13 http://security.ubuntu.com/ubuntu jammy-security/restricted
amd64 Packages [1,404 kB]
Get:14 http://security.ubuntu.com/ubuntu jammy-security/main amd64
Packages [1,202 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64
Packages [1,430 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64
Packages [1,275 kB]
Fetched 8,024 kB in 2s (4,428 kB/s)
Reading package lists... Done
Checking for apache2...
Installing apache2...
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libjansson4 liblua5.3-0
  mailcap
  mime-support ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-
  browser ufw
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libjansson4 liblua5.3-0
  mailcap
  mime-support ssl-cert
0 upgraded, 13 newly installed, 0 to remove and 8 not upgraded.
Need to get 2,135 kB of archives.
After this operation, 8,482 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64
```

```
libapr1 amd64 1.7.0-8ubuntu0.22.04.1 [108 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64
libaprutil1 amd64 1.6.1-5ubuntu4.22.04.2 [92.8 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64
libaprutil1-dbd-sqlite3 amd64 1.6.1-5ubuntu4.22.04.2 [11.3 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64
libaprutil1-ldap amd64 1.6.1-5ubuntu4.22.04.2 [9,170 B]
Get:5 http://archive.ubuntu.com/ubuntu jammy/main amd64 libjansson4
amd64 2.13.1-1.1build3 [32.4 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy/main amd64 liblua5.3-0
amd64 5.3.6-1build1 [140 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64
apache2-bin amd64 2.4.52-1ubuntu4.6 [1,345 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64
apache2-data all 2.4.52-1ubuntu4.6 [165 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64
apache2-utils amd64 2.4.52-1ubuntu4.6 [89.1 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy/main amd64 mailcap all
3.70+nmu1ubuntu1 [23.8 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy/main amd64 mime-support
all 3.66 [3,696 B]
Get:12 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64
apache2 amd64 2.4.52-1ubuntu4.6 [97.8 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy/main amd64 ssl-cert all
1.1.2 [17.4 kB]
Fetched 2,135 kB in 0s (9,239 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog
based frontend cannot be used. at
/usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 78, <> line 13.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (This frontend requires a controlling tty.)
debconf: falling back to frontend: Teletype
dpkg-preconfigure: unable to re-open stdin:
Selecting previously unselected package libapr1:amd64.
(Reading database ... 120882 files and directories currently
installed.)
Preparing to unpack .../00-libapr1_1.7.0-
8ubuntu0.22.04.1_amd64.deb ...
Unpacking libapr1:amd64 (1.7.0-8ubuntu0.22.04.1) ...
Selecting previously unselected package libaprutil1:amd64.
Preparing to unpack .../01-libaprutil1_1.6.1-
5ubuntu4.22.04.2_amd64.deb ...
Unpacking libaprutil1:amd64 (1.6.1-5ubuntu4.22.04.2) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../02-libaprutil1-dbd-sqlite3_1.6.1-
5ubuntu4.22.04.2_amd64.deb ...
Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.1-5ubuntu4.22.04.2) ...
```

```
Selecting previously unselected package libaprutil1-ldap:amd64.  
Preparing to unpack .../03-libaprutil1-ldap_1.6.1-  
5ubuntu4.22.04.2_amd64.deb ...  
Unpacking libaprutil1-ldap:amd64 (1.6.1-5ubuntu4.22.04.2) ...  
Selecting previously unselected package libjansson4:amd64.  
Preparing to unpack .../04-libjansson4_2.13.1-1.1build3_amd64.deb ...  
Unpacking libjansson4:amd64 (2.13.1-1.1build3) ...  
Selecting previously unselected package liblua5.3-0:amd64.  
Preparing to unpack .../05-liblua5.3-0_5.3.6-1build1_amd64.deb ...  
Unpacking liblua5.3-0:amd64 (5.3.6-1build1) ...  
Selecting previously unselected package apache2-bin.  
Preparing to unpack .../06-apache2-bin_2.4.52-1ubuntu4.6_amd64.deb ...  
Unpacking apache2-bin (2.4.52-1ubuntu4.6) ...  
Selecting previously unselected package apache2-data.  
Preparing to unpack .../07-apache2-data_2.4.52-1ubuntu4.6_all.deb ...  
Unpacking apache2-data (2.4.52-1ubuntu4.6) ...  
Selecting previously unselected package apache2-utils.  
Preparing to unpack .../08-apache2-utils_2.4.52-  
1ubuntu4.6_amd64.deb ...  
Unpacking apache2-utils (2.4.52-1ubuntu4.6) ...  
Selecting previously unselected package mailcap.  
Preparing to unpack .../09-mailcap_3.70+nmu1ubuntu1_all.deb ...  
Unpacking mailcap (3.70+nmu1ubuntu1) ...  
Selecting previously unselected package mime-support.  
Preparing to unpack .../10-mime-support_3.66_all.deb ...  
Unpacking mime-support (3.66) ...  
Selecting previously unselected package apache2.  
Preparing to unpack .../11-apache2_2.4.52-1ubuntu4.6_amd64.deb ...  
Unpacking apache2 (2.4.52-1ubuntu4.6) ...  
Selecting previously unselected package ssl-cert.  
Preparing to unpack .../12-ssl-cert_1.1.2_all.deb ...  
Unpacking ssl-cert (1.1.2) ...  
Setting up libapr1:amd64 (1.7.0-8ubuntu0.22.04.1) ...  
Setting up libjansson4:amd64 (2.13.1-1.1build3) ...  
Setting up ssl-cert (1.1.2) ...  
debconf: unable to initialize frontend: Dialog  
debconf: (No usable dialog-like program is installed, so the dialog  
based frontend cannot be used. at  
/usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 78.)  
debconf: falling back to frontend: Readline  
Setting up liblua5.3-0:amd64 (5.3.6-1build1) ...  
Setting up apache2-data (2.4.52-1ubuntu4.6) ...  
Setting up mailcap (3.70+nmu1ubuntu1) ...  
Setting up libaprutil1:amd64 (1.6.1-5ubuntu4.22.04.2) ...  
Setting up mime-support (3.66) ...  
Setting up libaprutil1-ldap:amd64 (1.6.1-5ubuntu4.22.04.2) ...  
Setting up libaprutil1-dbd-sqlite3:amd64 (1.6.1-5ubuntu4.22.04.2) ...  
Setting up apache2-utils (2.4.52-1ubuntu4.6) ...  
Setting up apache2-bin (2.4.52-1ubuntu4.6) ...
```

```
Setting up apache2 (2.4.52-1ubuntu4.6) ...
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of start.
Created symlink
/etc/systemd/system/multi-user.target.wants/apache2.service →
/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-
htcacheclean.service → /lib/systemd/system/apache-
htcacheclean.service.
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.4) ...
/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a
symbolic link
```

```
/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link
```

Enabling and starting Apache...

Synchronizing state of apache2.service with SysV service script with /lib/systemd/systemd-sysv-install.

Executing: /lib/systemd/systemd-sysv-install enable apache2

System has not been booted with systemd as init system (PID 1). Can't operate.

Failed to connect to bus: Host is down

An error occurred. Exiting...

```
%%writefile advanced_config_mgmt.sh
```

```
#!/bin/bash
```

```
# Function to check if a command was successful
```

```
check_success() {  
    if [ $? -ne 0 ]; then  
        echo "An error occurred. Exiting..."  
        exit 1  
    fi  
}
```

```
# Function to install a package if it's not already installed
```

```
install_package() {  
    echo "Checking for $1..."  
    if ! dpkg -l | grep -q "$1"; then  
        echo "Installing $1..."  
        sudo apt-get install -y $1  
        check_success  
    else  
        echo "$1 is already installed."  
    fi  
}
```

```
# Setting up environment variables
```

```
setup_environment() {  
    echo "Setting up environment variables..."  
    export MY_ENV_VAR="MyValue"  
    echo "Environment variable MY_ENV_VAR set to $MY_ENV_VAR"  
}
```

```
# Creating a new user (if not already exists)
```

```
create_user() {  
    local username=$1  
    echo "Creating a new user: $username"  
    if ! id "$username" &>/dev/null; then  
        sudo useradd -m -s /bin/bash "$username"  
    fi  
}
```



```

        check_success
        echo "User $username created."
    else
        echo "User $username already exists."
    fi
}

# Advanced package management
advanced_package_management() {
    echo "Adding a PPA and installing a package from it..."
    sudo add-apt-repository ppa:deadsnakes/ppa -y
    check_success
    install_package python3.8
}

# Setting up a Python environment
setup_python_environment() {
    echo "Setting up Python environment..."
    sudo apt-get install -y python3-pip python3-venv
    check_success
    python3 -m venv myenv
    source myenv/bin/activate
    pip install numpy pandas
    echo "Python environment set up with numpy and pandas installed."
}

# Main program
main() {
    echo "Starting advanced configuration management script..."
    sleep 2

    # Update package list
    echo "Updating package list..."
    sudo apt-get update
    check_success

    setup_environment
    create_user "newuser"
    advanced_package_management
    setup_python_environment

    echo "Configuration management tasks completed."
}

main

Writing advanced_config_mgmt.sh

!bash advanced_config_mgmt.sh

```

```
Starting advanced configuration management script...
Updating package list...
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [110
kB]
Hit:3 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/
InRelease
Get:4 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [119
kB]
Hit:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:6
https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/
x86_64 InRelease
Hit:7 https://ppa.launchpadcontent.net/c2d4u.team/c2d4u4.0+/ubuntu
jammy InRelease
Hit:8 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy
InRelease
Hit:9 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu
jammy InRelease
Hit:10 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy
InRelease
Fetched 229 kB in 1s (255 kB/s)
Reading package lists... Done
Setting up environment variables...
Environment variable MY_ENV_VAR set to MyValue
Creating a new user: newuser
User newuser created.
Adding a PPA and installing a package from it...
Repository: 'deb
https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu/ jammy main'
Description:
This PPA contains more recent Python versions packaged for Ubuntu.

Disclaimer: there's no guarantee of timely updates in case of security
problems or other issues. If you want to use them in a security-or-
otherwise-critical environment (say, on a production server), you do
so at your own risk.
```

Update Note

=====

Please use this repository instead of ppa:fkrull/deadsnakes.

Reporting Issues

=====

Issues can be reported in the master issue tracker at:
<https://github.com/deadsnakes/issues/issues>

Supported Ubuntu and Python Versions

=====

- Ubuntu 20.04 (focal) Python3.5 - Python3.7, Python3.9 - Python3.13
- Ubuntu 22.04 (jammy) Python3.7 - Python3.9, Python3.11 - Python3.13
- Note: Python2.7 (all), Python 3.8 (focal), Python 3.10 (jammy) are not provided by deadsnakes as upstream ubuntu provides those packages.

Why some packages aren't built:

- Note: for focal, older python versions require libssl<1.1 so they are not currently built
- Note: for jammy, older python versions require libssl<3 so they are not currently built
- If you need these, reach out to asottile to set up a private ppa

The packages may also work on other versions of Ubuntu or Debian, but that is not tested or supported.

Packages

=====

The packages provided here are loosely based on the debian upstream packages with some modifications to make them more usable as non-default pythons and on ubuntu. As such, the packages follow debian's patterns and often do not include a full python distribution with just ``apt install python#.#``. Here is a list of packages that may be useful along with the default install:

- ``python#.#-dev``: includes development headers for building C extensions
- ``python#.#-venv``: provides the standard library ``venv`` module
- ``python#.#-distutils``: provides the standard library ``distutils`` module
- ``python#.#-lib2to3``: provides the ``2to3-#.#`` utility as well as the standard library ``lib2to3`` module
- ``python#.#-gdbm``: provides the standard library ``dbm.gnu`` module
- ``python#.#-tk``: provides the standard library ``tkinter`` module

Third-Party Python Modules

=====

Python modules in the official Ubuntu repositories are packaged to work with the Python interpreters from the official repositories. Accordingly, they generally won't work with the Python interpreters from this PPA. As an exception, pure-Python modules for Python 3 will work, but any compiled extension modules won't.

To install 3rd-party Python modules, you should use the common Python packaging tools. For an introduction into the Python packaging ecosystem and its tools, refer to the Python Packaging User Guide: <https://packaging.python.org/installing/>

Sources

=====

The package sources are available at:
<https://github.com/deadsnakes/>

Nightly Builds

=====

```
For nightly builds, see ppa:deadsnakes/nightly
https://launchpad.net/~deadsnakes/+archive/ubuntu/nightly
More info: https://launchpad.net/~deadsnakes/+archive/ubuntu/ppa
Adding repository.
Found existing deb entry in /etc/apt/sources.list.d/deadsnakes-ubuntu-
ppa-jammy.list
Adding deb entry to /etc/apt/sources.list.d/deadsnakes-ubuntu-ppa-
jammy.list
Found existing deb-src entry in /etc/apt/sources.list.d/deadsnakes-
ubuntu-ppa-jammy.list
Adding disabled deb-src entry to /etc/apt/sources.list.d/deadsnakes-
ubuntu-ppa-jammy.list
Adding key to /etc/apt/trusted.gpg.d/deadsnakes-ubuntu-ppa.gpg with
fingerprint F23C5A6CF475977595C89F51BA6932366A755776
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/
InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:5
https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/
x86_64 InRelease
Hit:6 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:7 https://ppa.launchpadcontent.net/c2d4u.team/c2d4u4.0+/ubuntu
jammy InRelease
Hit:8 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy
InRelease
Hit:9 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu
jammy InRelease
Hit:10 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy
InRelease
Reading package lists... Done
Checking for python3.8...
Installing python3.8...
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libpython3.8-minimal libpython3.8-stdlib python3.8-minimal
Suggested packages:
  python3.8-venv binfmt-support
The following NEW packages will be installed:
  libpython3.8-minimal libpython3.8-stdlib python3.8 python3.8-minimal
```

```
0 upgraded, 4 newly installed, 0 to remove and 8 not upgraded.
Need to get 5,071 kB of archives.
After this operation, 18.8 MB of additional disk space will be used.
Get:1 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu
jammy/main amd64 libpython3.8-minimal amd64 3.8.18-1+jammy1 [794 kB]
Get:2 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu
jammy/main amd64 python3.8-minimal amd64 3.8.18-1+jammy1 [2,024 kB]
Get:3 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu
jammy/main amd64 libpython3.8-stdlib amd64 3.8.18-1+jammy1 [1,815 kB]
Get:4 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu
jammy/main amd64 python3.8 amd64 3.8.18-1+jammy1 [438 kB]
Fetched 5,071 kB in 5s (1,002 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog
based frontend cannot be used. at
/usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 78, <> line 4.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (This frontend requires a controlling tty.)
debconf: falling back to frontend: Teletype
dpkg-preconfigure: unable to re-open stdin:
Selecting previously unselected package libpython3.8-minimal:amd64.
(Reading database ... 121634 files and directories currently
installed.)
Preparing to unpack .../libpython3.8-minimal_3.8.18-1+jammy1_amd64.deb
...
Unpacking libpython3.8-minimal:amd64 (3.8.18-1+jammy1) ...
Selecting previously unselected package python3.8-minimal.
Preparing to unpack .../python3.8-minimal_3.8.18-
1+jammy1_amd64.deb ...
Unpacking python3.8-minimal (3.8.18-1+jammy1) ...
Selecting previously unselected package libpython3.8-stdlib:amd64.
Preparing to unpack .../libpython3.8-stdlib_3.8.18-
1+jammy1_amd64.deb ...
Unpacking libpython3.8-stdlib:amd64 (3.8.18-1+jammy1) ...
Selecting previously unselected package python3.8.
Preparing to unpack .../python3.8_3.8.18-1+jammy1_amd64.deb ...
Unpacking python3.8 (3.8.18-1+jammy1) ...
Setting up libpython3.8-minimal:amd64 (3.8.18-1+jammy1) ...
Setting up python3.8-minimal (3.8.18-1+jammy1) ...
Setting up libpython3.8-stdlib:amd64 (3.8.18-1+jammy1) ...
Setting up python3.8 (3.8.18-1+jammy1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmulubuntu1) ...
Setting up Python environment...
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
python3-pip-whl python3-setuptools python3-setuptools-whl python3-
```

```
wheel
python3.10-venv
Suggested packages:
python-setuptools-doc
The following NEW packages will be installed:
python3-pip python3-pip-whl python3-setuptools python3-setuptools-
whl
python3-venv python3-wheel python3.10-venv
0 upgraded, 7 newly installed, 0 to remove and 8 not upgraded.
Need to get 4,151 kB of archives.
After this operation, 11.9 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64
python3-setuptools all 59.6.0-1.2ubuntu0.22.04.1 [339 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64
python3-wheel all 0.37.1-2ubuntu0.22.04.1 [32.0 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64
python3-pip all 22.0.2+dfsg-1ubuntu0.3 [1,305 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64
python3-pip-whl all 22.0.2+dfsg-1ubuntu0.3 [1,679 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64
python3-setuptools-whl all 59.6.0-1.2ubuntu0.22.04.1 [788 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64
python3.10-venv amd64 3.10.12-1~22.04.2 [5,724 B]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64
python3-venv amd64 3.10.6-1~22.04 [1,038 B]
Fetched 4,151 kB in 0s (21.3 MB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog
based frontend cannot be used. at
/usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 78, <> line 7.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (This frontend requires a controlling tty.)
debconf: falling back to frontend: Teletype
dpkg-preconfigure: unable to re-open stdin:
Selecting previously unselected package python3-setuptools.
(Reading database ... 122260 files and directories currently
installed.)
Preparing to unpack .../0-python3-setuptools_59.6.0-
1.2ubuntu0.22.04.1_all.deb ...
Unpacking python3-setuptools (59.6.0-1.2ubuntu0.22.04.1) ...
Selecting previously unselected package python3-wheel.
Preparing to unpack .../1-python3-wheel_0.37.1-
2ubuntu0.22.04.1_all.deb ...
Unpacking python3-wheel (0.37.1-2ubuntu0.22.04.1) ...
Selecting previously unselected package python3-pip.
Preparing to unpack .../2-python3-pip_22.0.2+dfsg-
1ubuntu0.3_all.deb ...
Unpacking python3-pip (22.0.2+dfsg-1ubuntu0.3) ...
Selecting previously unselected package python3-pip-whl.
```

```

Preparing to unpack .../3-python3-pip-whl_22.0.2+dfsg-
1ubuntu0.3_all.deb ...
Unpacking python3-pip-whl (22.0.2+dfsg-1ubuntu0.3) ...
Selecting previously unselected package python3-setuptools-whl.
Preparing to unpack .../4-python3-setuptools-whl_59.6.0-
1.2ubuntu0.22.04.1_all.deb ...
Unpacking python3-setuptools-whl (59.6.0-1.2ubuntu0.22.04.1) ...
Selecting previously unselected package python3.10-venv.
Preparing to unpack .../5-python3.10-venv_3.10.12-
1~22.04.2_amd64.deb ...
Unpacking python3.10-venv (3.10.12-1~22.04.2) ...
Selecting previously unselected package python3-venv.
Preparing to unpack .../6-python3-venv_3.10.6-1~22.04_amd64.deb ...
Unpacking python3-venv (3.10.6-1~22.04) ...
Setting up python3-setuptools-whl (59.6.0-1.2ubuntu0.22.04.1) ...
Setting up python3-setuptools (59.6.0-1.2ubuntu0.22.04.1) ...
Setting up python3-pip-whl (22.0.2+dfsg-1ubuntu0.3) ...
Setting up python3-wheel (0.37.1-2ubuntu0.22.04.1) ...
Setting up python3-pip (22.0.2+dfsg-1ubuntu0.3) ...
Setting up python3.10-venv (3.10.12-1~22.04.2) ...
Setting up python3-venv (3.10.6-1~22.04) ...
Processing triggers for man-db (2.10.2-1) ...
Collecting numpy
  Downloading numpy-1.26.2-cp310-cp310-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl (18.2 MB)
  _____ 18.2/18.2 MB 66.5 MB/s eta
0:00:00
anylinux_2_17_x86_64.manylinux2014_x86_64.whl (12.3 MB)
  _____ 12.3/12.3 MB 98.2 MB/s eta
0:00:00
  _____ 502.5/502.5 KB 49.4 MB/s eta
0:00:00
  _____ 247.7/247.7 KB 25.9 MB/s eta
0:00:00
  _____ 341.8/341.8 KB 38.9 MB/s eta
0:00:00
py, python-dateutil, pandas
Successfully installed numpy-1.26.2 pandas-2.1.3 python-dateutil-2.8.2
pytz-2023.3.post1 six-1.16.0 tzdata-2023.3
Python environment set up with numpy and pandas installed.
Configuration management tasks completed.

%%writefile advanced_config_mgmt_ver2.sh

#!/bin/bash

# Function to check if a command was successful
check_success() {
    if [ $? -ne 0 ]; then
        echo "An error occurred. Exiting..."
    fi
}

```

```

        exit 1
    fi
}

# Function to install a package if it's not already installed
install_package() {
    echo "Checking for $1..."
    if ! dpkg -l | grep -q "$1"; then
        echo "Installing $1..."
        sudo apt-get install -y $1
        check_success
    else
        echo "$1 is already installed."
    fi
}

# Setting up environment variables
setup_environment() {
    echo "Setting up environment variables..."
    export MY_ENV_VAR="MyValue"
    echo "Environment variable MY_ENV_VAR set to $MY_ENV_VAR"
}

# Creating a new user (if not already exists)
create_user() {
    local username=$1
    echo "Creating a new user: $username"
    if ! id "$username" &>/dev/null; then
        sudo useradd -m -s /bin/bash "$username"
        check_success
        echo "User $username created."
    else
        echo "User $username already exists."
    fi
}

# Advanced package management
advanced_package_management() {
    echo "Adding a PPA and installing a package from it..."
    sudo add-apt-repository ppa:deadsnakes/ppa -y
    check_success
    install_package python3.8
}

# Setting up a Python environment
setup_python_environment() {
    echo "Setting up Python environment..."
    sudo apt-get install -y python3-pip python3-venv
    check_success
    python3 -m venv myenv

```



```

    source myenv/bin/activate
    pip install numpy pandas
    echo "Python environment set up with numpy and pandas installed."
}

# Set up a cron job
setup_cron_job() {
    echo "Setting up a cron job to run a script daily..."
    # Write out current crontab
    crontab -l > mycron
    # Echo new cron into cron file
    echo "0 0 * * * /path/to/daily_task.sh" >> mycron
    # Install new cron file
    crontab mycron
    rm mycron
    echo "Cron job set up."
}

# Cloning a repository from GitHub
clone_repository() {
    echo "Cloning a repository from GitHub..."
    git clone https://github.com/exampleuser/example-repo.git
    check_success
    echo "Repository cloned."
}

# Creating a symbolic link
create_symbolic_link() {
    echo "Creating a symbolic link..."
    ln -s /path/to/file /path/to/symlink
    check_success
    echo "Symbolic link created."
}

# Main program
main() {
    echo "Starting advanced configuration management script..."
    sleep 2

    # Update package list
    echo "Updating package list..."
    sudo apt-get update
    check_success

    setup_environment
    create_user "newuser"
    advanced_package_management
    setup_python_environment
    setup_cron_job
    clone_repository

```

```

    create_symbolic_link
    echo "Configuration management tasks completed."
}

main

Writing advanced_config_mgmt_ver2.sh

!bash advanced_config_mgmt_ver2.sh

Starting advanced configuration management script...
Updating package list...
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:5 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/
InRelease
Hit:6
https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/
x86_64 InRelease
Hit:7 https://ppa.launchpadcontent.net/c2d4u.team/c2d4u4.0+/ubuntu
jammy InRelease
Hit:8 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy
InRelease
Hit:9 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu
jammy InRelease
Hit:10 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy
InRelease
Reading package lists... Done
Setting up environment variables...
Environment variable MY_ENV_VAR set to MyValue
Creating a new user: newuser
User newuser already exists.
Adding a PPA and installing a package from it...
Repository: 'deb
https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu/ jammy main'
Description:
This PPA contains more recent Python versions packaged for Ubuntu.

Disclaimer: there's no guarantee of timely updates in case of security
problems or other issues. If you want to use them in a security-or-
otherwise-critical environment (say, on a production server), you do
so at your own risk.

Update Note
=====
Please use this repository instead of ppa:fkrull/deadsnakes.

```

Reporting Issues

=====

Issues can be reported in the master issue tracker at:
<https://github.com/deadsnakes/issues/issues>

Supported Ubuntu and Python Versions

=====

- Ubuntu 20.04 (focal) Python3.5 - Python3.7, Python3.9 - Python3.13
- Ubuntu 22.04 (jammy) Python3.7 - Python3.9, Python3.11 - Python3.13
- Note: Python2.7 (all), Python 3.8 (focal), Python 3.10 (jammy) are not provided by deadsnakes as upstream ubuntu provides those packages.

Why some packages aren't built:

- Note: for focal, older python versions require libssl<1.1 so they are not currently built
- Note: for jammy, older python versions require libssl<3 so they are not currently built
- If you need these, reach out to asottile to set up a private ppa

The packages may also work on other versions of Ubuntu or Debian, but that is not tested or supported.

Packages

=====

The packages provided here are loosely based on the debian upstream packages with some modifications to make them more usable as non-default pythons and on ubuntu. As such, the packages follow debian's patterns and often do not include a full python distribution with just ``apt install python#.#``. Here is a list of packages that may be useful along with the default install:

- ``python#.#-dev``: includes development headers for building C extensions
- ``python#.#-venv``: provides the standard library ``venv`` module
- ``python#.#-distutils``: provides the standard library ``distutils`` module
- ``python#.#-lib2to3``: provides the ``2to3-#.#`` utility as well as the standard library ``lib2to3`` module
- ``python#.#-gdbm``: provides the standard library ``dbm.gnu`` module
- ``python#.#-tk``: provides the standard library ``tkinter`` module

Third-Party Python Modules

=====

Python modules in the official Ubuntu repositories are packaged to work with the Python interpreters from the official repositories. Accordingly, they generally won't work with the Python interpreters

from this PPA. As an exception, pure-Python modules for Python 3 will work, but any compiled extension modules won't.

To install 3rd-party Python modules, you should use the common Python packaging tools. For an introduction into the Python packaging ecosystem and its tools, refer to the Python Packaging User Guide: <https://packaging.python.org/installing/>

Sources

=====

The package sources are available at:

<https://github.com/deadsnakes/>

Nightly Builds

=====

For nightly builds, see [ppa:deadsnakes/nightly](https://launchpad.net/~deadsnakes/nightly)

<https://launchpad.net/~deadsnakes/+archive/ubuntu/nightly>

More info: <https://launchpad.net/~deadsnakes/+archive/ubuntu/ppa>

Adding repository.

Found existing deb entry in /etc/apt/sources.list.d/deadsnakes-ubuntu-ppa-jammy.list

Adding deb entry to /etc/apt/sources.list.d/deadsnakes-ubuntu-ppa-jammy.list

Found existing deb-src entry in /etc/apt/sources.list.d/deadsnakes-ubuntu-ppa-jammy.list

Adding disabled deb-src entry to /etc/apt/sources.list.d/deadsnakes-ubuntu-ppa-jammy.list

Adding key to /etc/apt/trusted.gpg.d/deadsnakes-ubuntu-ppa.gpg with fingerprint F23C5A6CF475977595C89F51BA6932366A755776

Hit:1 <https://cloud.r-project.org/bin/linux/ubuntu> jammy-cran40/InRelease

Hit:2

https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64 InRelease

Hit:3 <http://security.ubuntu.com/ubuntu> jammy-security InRelease

Hit:4 <http://archive.ubuntu.com/ubuntu> jammy InRelease

Hit:5 <http://archive.ubuntu.com/ubuntu> jammy-updates InRelease

Hit:6 <http://archive.ubuntu.com/ubuntu> jammy-backports InRelease

Hit:7 <https://ppa.launchpadcontent.net/c2d4u.team/c2d4u4.0+/ubuntu> jammy InRelease

Hit:8 <https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu> jammy InRelease

Hit:9 <https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu> jammy InRelease

Hit:10 <https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu> jammy InRelease

Reading package lists... Done

Checking for python3.8...

python3.8 is already installed.

```
Setting up Python environment...
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3-pip is already the newest version (22.0.2+dfsg-1ubuntu0.3).
python3-venv is already the newest version (3.10.6-1~22.04).
0 upgraded, 0 newly installed, 0 to remove and 8 not upgraded.
Requirement already satisfied: numpy in ./myenv/lib/python3.10/site-
packages (1.26.2)
Requirement already satisfied: pandas in ./myenv/lib/python3.10/site-
packages (2.1.3)
Requirement already satisfied: pytz>=2020.1 in
./myenv/lib/python3.10/site-packages (from pandas) (2023.3.post1)
Requirement already satisfied: python-dateutil>=2.8.2 in
./myenv/lib/python3.10/site-packages (from pandas) (2.8.2)
Requirement already satisfied: tzdata>=2022.1 in
./myenv/lib/python3.10/site-packages (from pandas) (2023.3)
Requirement already satisfied: six>=1.5 in
./myenv/lib/python3.10/site-packages (from python-dateutil>=2.8.2-
>pandas) (1.16.0)
Python environment set up with numpy and pandas installed.
Setting up a cron job to run a script daily...
advanced_config_mgmt_ver2.sh: line 67: crontab: command not found
advanced_config_mgmt_ver2.sh: line 71: crontab: command not found
Cron job set up.
Cloning a repository from GitHub...
Cloning into 'example-repo'...
fatal: could not read Username for 'https://github.com': No such
device or address
An error occurred. Exiting...
```